MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 1602.2
STANDARD	
Composite Imaging Local Set	**02 March 2023**

1 Scope

This standard defines a KLV metadata Local Set (LS) which enables a unique assignment of image-attribute metadata to several images composited into one image frame.

The standard supports image compositing such as: tiling of views from different sensors, blending of images, a stacked series of images, and picture-in-picture.

The standard applies to rectangular geometric images only; that is, rectangles-on-rectangles.

The standard assumes reuse of the Segment Local Set (see MISB ST 1607 [1]) within an instantiating metadata set.

The standard does not support dynamic overlays under direction of a client receiver.

2 References

[1] MISB ST 1607.1 Constructs to Amend/Segment KLV Metadata, Mar 2023.
[2] MISB ST 0601.19 UAS Datalink Local Set, Mar 2023.
[3] MISB ST 0107.5 KLV Metadata in Motion Imagery, Oct 2021.
[4] SMPTE ST 336:2017 Data Encoding Protocol Using Key-Length-Value.
[5] MISB ST 0807.26 MISB KLV Metadata Registry, Jun 2022.
[6] MISB ST 0603.5 MISP Time System and Timestamps, Oct 2017.
[7] MISB ST 1204.3 Motion Imagery Identification System (MIIS) Core Identifier, Feb 2020.
[8] MISB ST 0902.8 Motion Imagery Sensor Minimum Metadata Set, Nov 2018.
[9] MISB MISP-2023.2 Motion Imagery Standards Profile, Mar 2023.
[10] MISB ST 1202.3 Generalized Transformation Parameters, Mar 2023.

3 Acronyms and Definitions

KLV	Key-Length-Value
LS	Local Set
MISB	Motion Imagery Standards Board
MISP	Motion Imagery Standards Profile

MSID	Metadata Substream Identifier
PIP	Picture-in-Picture
SMPTE	Society of Motion Picture & Television Engineers
ST	Standard

4 Revision History

Revision	Date	Summary of Changes
ST 1602.2	03/02/2023	• Added Section 8.2 Use of ST 1607 and the Metadata Substream Identifier (MSID) • Changed Document Version Item to uint • Removed Keys from tables consistent with current practice • Updated Examples in the Appendices • Updated references

5 Introduction

Image composition, as used in this document, is an arrangement of one or more images – possibly but not necessarily – derived from one or more sensors, combined into one image of Motion Imagery. Examples include tiling of views from different sensors, blending of images, a stacked series of images, and picture-in-picture.

Whereas a single sensor source typically produces a Motion Imagery sequence, the metadata to describe an image composed of more than one image (called sub-images) is more complex in structure. Each sub-image within the composite requires description of its unique attributes, such as its position with the composited image, the pixel density, the source for the sub-image, etc. This necessitates a distribution of metadata among the full composite and the various sub-images, in addition to each sub-image specific attribute metadata (i.e., position, pixel density, etc.).

The Composite Imaging Local Set provides this second category of image-specific attribute metadata. In typical use cases, the Composite Imaging LS will be embedded within the Segment LS described by MISB ST 1607 [1], which affords multiple instances of the Composite Imaging LS within one instantiating metadata set (e.g., MISB ST 0601 [2]). Refer to the appendix for example use cases.

6 Background

This section provides terminology and an orientation for proper application of the metadata items within the Composite Imaging Local Set.

Composite Image: an image composed of several images derived from one or more sources of images within Motion Imagery.

Source Image: an image as produced by a Motion Imagery source, such as a sensor or camera.

Source Image AOI: Area of Interest, which represents a rectangular area with a Source Image used in defining a Sub-Image.

Sub-Image: an image derived from a Source Image. A Sub-Image may represent a processed version of its Source Image (i.e., same pixel density but different image characteristics), or an area of interest extracted from a Source Image, also potentially processed.

The pixel density of a Composite Image is independent of Source Images represented. Consider the Composite Image a canvas for mapping Source Images. Figure 1 illustrates a generalized representation of the mapping process.

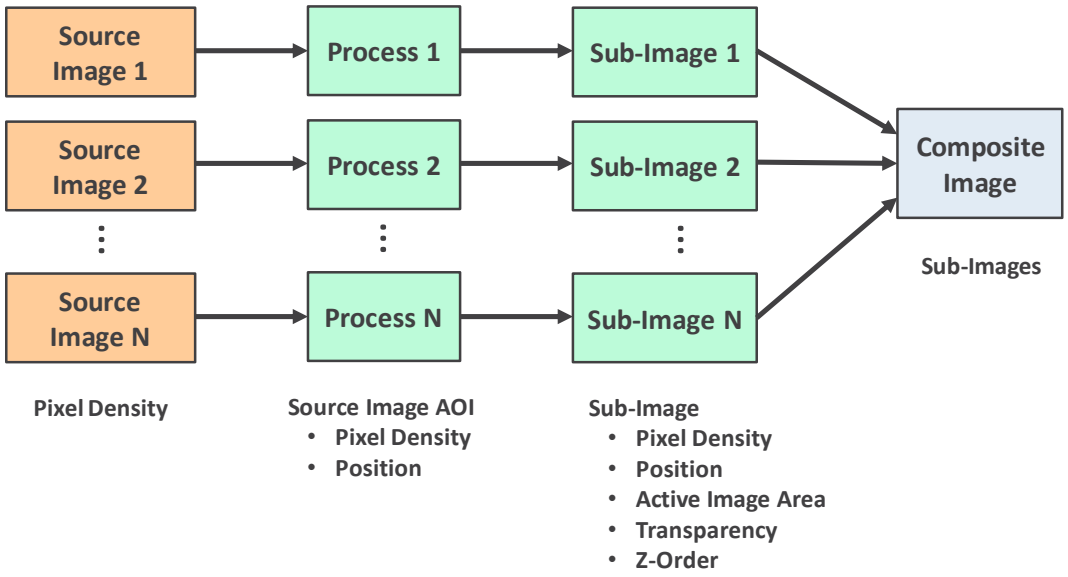


Figure 1: General mapping of multiple Source Images to Composite Image

Each Source Image in Figure 1 can undergo some process, such as color space conversion, bit-depth remapping, pixel density scaling, etc. prior to its placement within the Composite Image. Several parameters, discussed in later sections, complete the mapping of a Source Image to the Composite Image. These include: A Source Image pixel's density specified as its number of rows and columns; an optional specification of the Source Image AOI rectangular region within the Source Image; and the Sub-Image produced using the Source Image AOI. A Sub-Image has a pixel density (i.e., rows, columns), its position within the Composite Image (i.e., position with respect to the upper left corner of the Composite Image), and optionally, an active area representing image content within the Sub-Image. A Sub-Image is qualified by its degree of transparency and its z-axis ordering.

Some items within the Composite Imaging LS are mandatory; others depend on whether they are relevant to describing the mapping of a Source Image to the Composite Image.

7 Composite Imaging Local Set

The Composite Imaging LS is a KLV LS construct whose items provide information specific to an image, such as position, pixel density, transparency, and Z-order. All KLV metadata needs to conform to MISB ST 0107 [3] and SMPTE 336 [4] encoding rules.

7.1 Conventions

Requirement(s)	
ST 1602-01	All KLV metadata shall be expressed in accordance with MISB ST 0107.
ST 1602-02	Formatting of Composite Imaging metadata shall be compliant with SMPTE ST 336 encoding rules for Universal Labels and Local Sets.

7.2 Composite Imaging LS

The Composite Imaging LS 16-Byte Universal Label registered in MISB ST 0807 [5] is:

06.0E.2B.34.02.0B.01.01.0E.01.03.03.02.00.00.00 (CRC 666)

Table 1 summarizes the Composite Imaging LS with the following column definitions:

- **Tag** indicates the Tag to use when specifying the item in the Composite Imaging LS
- **Name** is the name of the dictionary item
- **Key** is the KLV registry Universal Label associated with the item
- **Units** specifies the units of an item's value
- **Type** specifies the data type(binary format) when encoding the value into binary; the Types uint and int allow the application to define the number of bytes needed for a values representation with the length indicated by the "Length" of the KLV triplet
- **Rules** indicates if the item is Mandatory or Optional; the Local Set must include items listed as Mandatory, while those listed as Optional are at the implementer's discretion

Table 1: Composite Imaging Local Set

Tag	Name	Units	Type	Rules
1	Precision Time Stamp	µs	uint64	OPTIONAL
2	Document Version	N/A	uint	MANDATORY
3	Source Image Rows	pixels	uint	OPTIONAL
4	Source Image Columns	pixels	uint	OPTIONAL
5	Source Image AOI Rows	pixels	uint	OPTIONAL
6	Source Image AOI Columns	pixels	uint	OPTIONAL
7	Source Image AOI Position X	pixels	int	OPTIONAL
8	Source Image AOI Position Y	pixels	int	OPTIONAL
9	Sub-Image Rows	pixels	uint	MANDATORY
10	Sub-Image Columns	pixels	uint	MANDATORY
11	Sub-Image Position X	pixels	int	MANDATORY
12	Sub-Image Position Y	pixels	int	MANDATORY

13	Active Sub-Image Rows	pixels	uint	OPTIONAL
14	Active Sub-Image Columns	pixels	uint	OPTIONAL
15	Active Sub-Image Offset X	pixels	int	OPTIONAL
16	Active Sub-Image Offset Y	pixels	int	OPTIONAL
17	Transparency	N/A	uint8	OPTIONAL
18	Z-Order	N/A	uint8	MANDATORY

7.3 Item Details

7.3.1 Precision Time Stamp

The Precision Time Stamp is the time of the composited image. The MISP mandates this item within a Motion Imagery stream. If the composited image's Precision Time Stamp is the same as the parent's Precision Time Stamp, it is optional in the Composite Imaging Local Set. Otherwise, provide the Precision Time Stamp in the Composite Imaging LS (e.g., in cases where a Source Image, or subsequent process, has a different timestamp than the parent timestamp). The use case will determine if additional timestamps are important to the application.

MISB ST 0603 [6] defines the Precision Time Stamp.

7.3.2 Document Version

The Document Version (Item 2) identifies the version of ST 1602 used in the implementation.

Requirement	
ST 1602-03	A Composite Imaging Local Set shall contain the Document Version (Item 2) metadata item.

7.3.3 Source Image

The Source Image is the image directly produced by the sensor imager. The Source Image Rows and Source Image Columns follow from the number of samples in the vertical and horizontal dimensions, respectively.

7.3.3.1 Source Image Rows

Source Image Rows (Item 3) is the number of image samples in the vertical direction (i.e., image height) of the source image for the described sub-image. See Figure 2.



Figure 2: Source Image pixel density specified as Rows, Columns

7.3.3.2 Source Image Columns

Source Image Columns (Item 4) is the number of image samples in the horizontal direction (i.e., image width) of the source image for the described sub-image. See Figure 2.

7.3.4 Source Image AOI

The Source Image AOI is a rectangular area-of-interest extracted from the Source Image. The following parameters describe the number of samples vertically (Source Image AOI Rows), the number of samples horizontally (Source Image AOI Columns). The Source Image AOI X Position and Source Image AOI Y Position parameters describe the offset position of the AOI relative to the Source Image AOI position of (0,0).

7.3.4.1 Source Image AOI Rows

Source Image AOI Rows (Item 5) is the number of image samples in the vertical direction (i.e., image height) of the Source Image AOI. The Source Image AOI is the image used to create a Sub-Image of the Source Image. See Figure 3. In cases where the entire Source Image maps directly to its Sub-Image (i.e., Source Image AOI identical to the Source Image), this parameter is not necessary to specify.

7.3.4.2 Source Image AOI Columns

Source Image AOI Columns (Item 6) is the number of image samples in the horizontal direction (i.e., image width) of the Source Image AOI. The Source Image AOI is the image used to create a Sub-Image of the Source Image. See Figure 3. In cases where the entire Source Image maps directly to its Sub-Image (i.e., Source Image AOI identical to the Source Image), this parameter is not necessary to specify.

7.3.4.3 Source Image AOI X Position

Source Image AOI X Position (Item 7) is the x-coordinate in pixels defining the upper left corner of the rectangular area of interest in the Source Image referenced to the upper left (0, 0) corner of the Source Image. The Source Image AOI is the image used to create a Sub-Image of the Source Image. See Figure 3. In cases where the entire Source Image maps directly to its Sub-Image (i.e., Source Image AOI identical to the Source Image), this parameter is not necessary to specify.

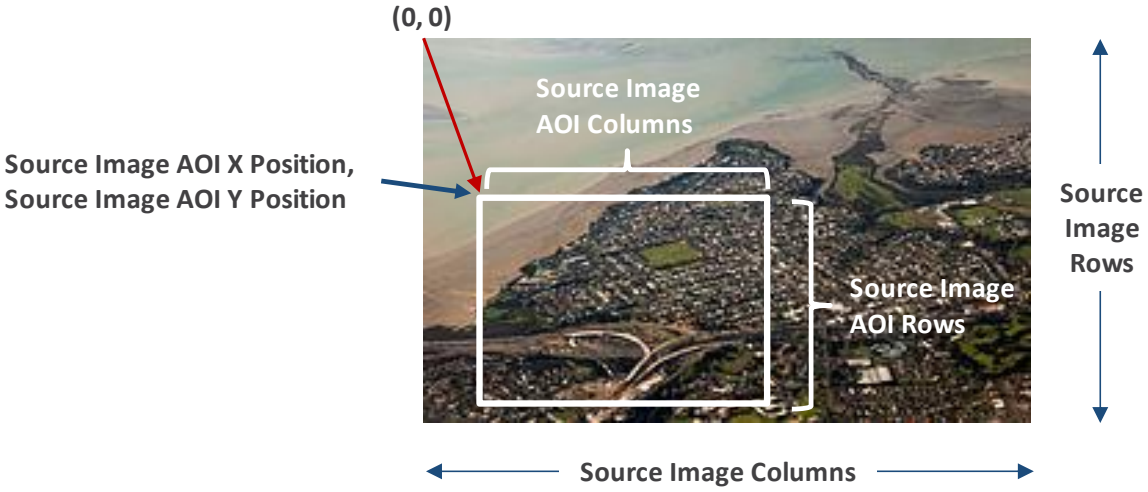


Figure 3: Source Image AOI position referenced to Source Image

7.3.4.4 Source Image AOI Y Position

Source Image AOI Y Position (Item 8) is the y-coordinate in pixels defining the upper left corner of the rectangular area of interest in the Source Image referenced to the upper left (0, 0) corner of the Source Image. The Source Image AOI is the image used to create a Sub-Image of the Source Image. See Figure 3. In cases where the entire Source Image maps directly to its Sub-Image (i.e., Source Image AOI identical to the Source Image), this parameter is not necessary to specify.

7.3.5 Sub-Image

A Sub-Image may represent a processed version of its Source Image (i.e., same pixel density but different image characteristics), or an area of interest extracted from a Source Image, also possibly processed. The Sub-Image Columns and Sub-Image Rows parameters specify the dimensions of the Sub-Image. The position of the Sub-Image is with respect to the (0,0) location of composite image created; the Sub-Image Position X and Sub-Image Position Y parameters specify this offset.

In the event a Sub-Image does not contain all active pixels, the Active Sub-Image Rows and Active Sub-Image Columns parameters enable specifying the rectangular region containing only active image samples. The parameters Active Sub-Image Offset X and Active Sub-Image Offset Y provide position information of where the active image lies with respect the Sub-Image.

7.3.5.1 Sub-Image Rows

Sub-Image Rows (Item 9) is the number of image samples in the vertical direction (i.e., image height) of the sub-image derived from its Source Image AOI. See Figure 4.

Requirement	
ST 1602.1-05	A Composite Imaging Local Set shall contain the Sub-Image Rows (Item 9) metadata item.



Figure 4: Source Image AOI used to create Sub-Image

7.3.5.2 Sub-Image Columns

Sub-Image Columns (Item 10) is the number of image samples in the horizontal direction (i.e., image width) of the sub-image derived from its source image. See Figure 4.

Requirement	
ST 1602.1-06	A Composite Imaging Local Set shall contain the Sub-Image Columns (Item 10) metadata item.

7.3.5.3 Sub-Image Position X

Sub-Image Position X (Item 11) is the x-coordinate of the upper left corner of a Sub-Image referenced to the upper left corner, i.e., coordinates (0, 0) of the Composite Image. See Figure 5.

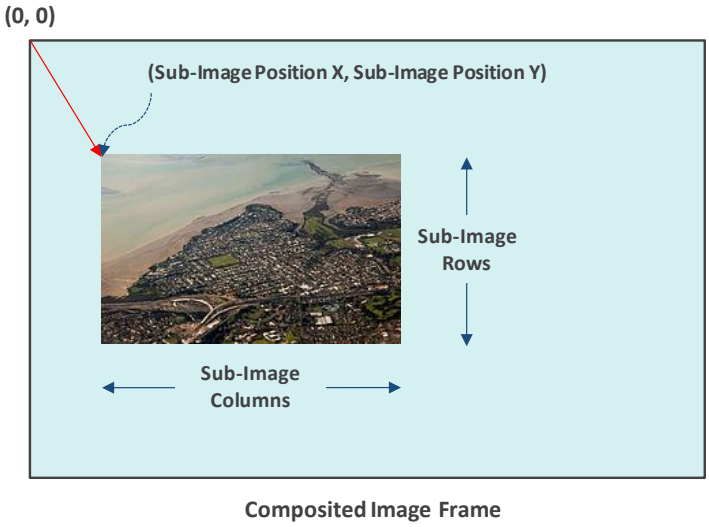


Figure 5: Relation of Sub-Image to Composite Image

	Requirement
ST 1602.1-07	A Composite Imaging Local Set shall contain the Sub-Image Position X (Item 11) metadata item.

7.3.5.4 Sub-Image Position Y

Sub-Image Position Y (Item 12) is the y-coordinate of the upper left corner of a Sub-Image referenced to the upper left corner, i.e., coordinates (0, 0) of the Composite Image. See Figure 5.

	Requirement
ST 1602.1-08	A Composite Imaging Local Set shall contain the Sub-Image Position Y (Item 12) metadata item.

7.3.5.5 Active Sub-Image Rows

Active Sub-Image Rows (Item 13) is the number of samples in the vertical direction representing Source Image content; that is, containing the portion of the Source Image of interest. See Figure 6.

7.3.5.6 Active Sub-Image Columns

Active Sub-Image Columns (Item 14) is the number of samples in the horizontal direction representing Source Image content; that is, containing the portion of the Source Image of interest. See Figure 6.

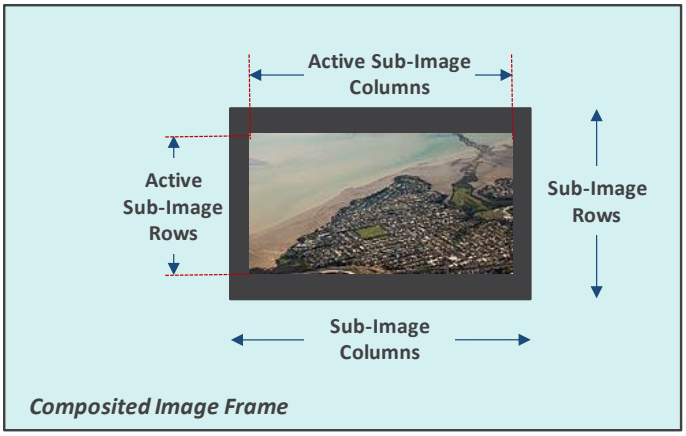


Figure 6: Active Sub-Image Area

7.3.5.7 Active Sub-Image Offset X

Active Sub-Image Offset X (Item 15) provides the x-coordinate of the upper left corner of the active image within the Sub-Image referenced to Sub-Image Position X. See Figure 7, where x-coordinate = Sub-Image Position X + Active Sub-Image Position X. Valid range: values greater than zero (0).

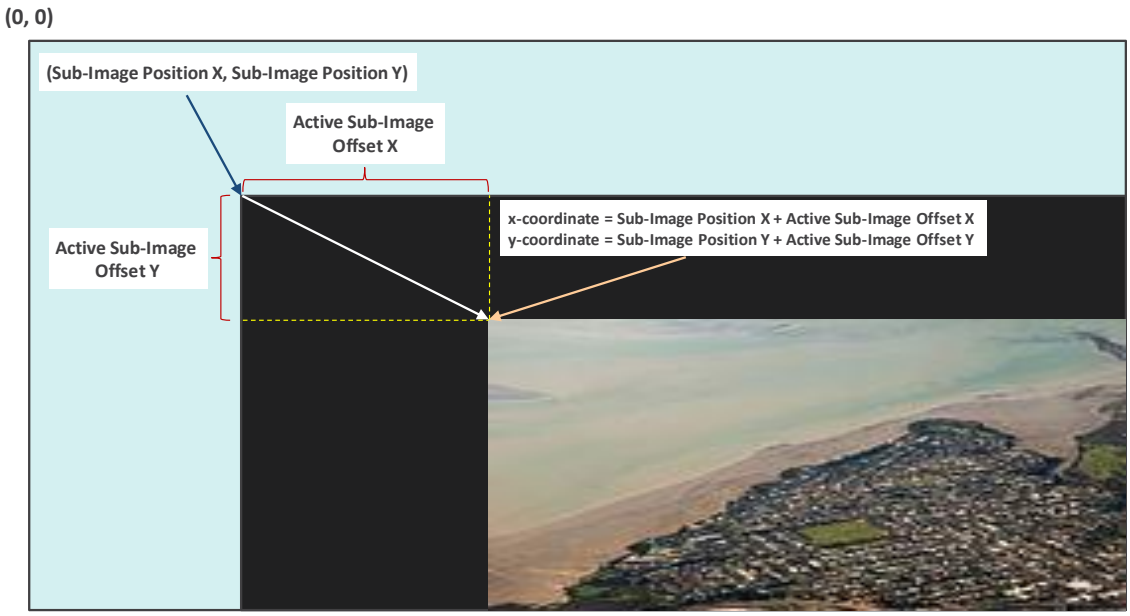


Figure 7: Relation of Active Sub-Image to Sub-Image

Knowledge of the Sub-Image Rows/Columns, Sub-Image Position X/Y, Active Image Rows/Columns, and Active Sub-Image Offset X/Y allows identifying non-image areas, such as black borders when an image is letter boxed.

7.3.5.8 Active Sub-Image Offset Y

Active Sub-Image Offset Y (Item 16) provides the y-coordinate of the upper left corner of the active image within the Sub-Image described by the Local Set referenced to Sub-Image Position Y. See Figure 7, where y-coordinate = Sub-Image Position Y + Active Sub-Image Position Y. Valid range: values greater than zero (0).

Use the Sub-Image Rows/Columns, Sub-Image Position X/Y, Active Sub-Image Rows/Columns, and Active Sub-Image Offset X/Y to describe non-image areas, such as black borders when an image is letter boxed.

7.3.6 Transparency

The Transparency (Item 17) parameter provides 256 levels of image transparency. Consider two images A and B superimposed on top of one another as shown in Figure 8; denote image A as the background image and image B as the foreground image.

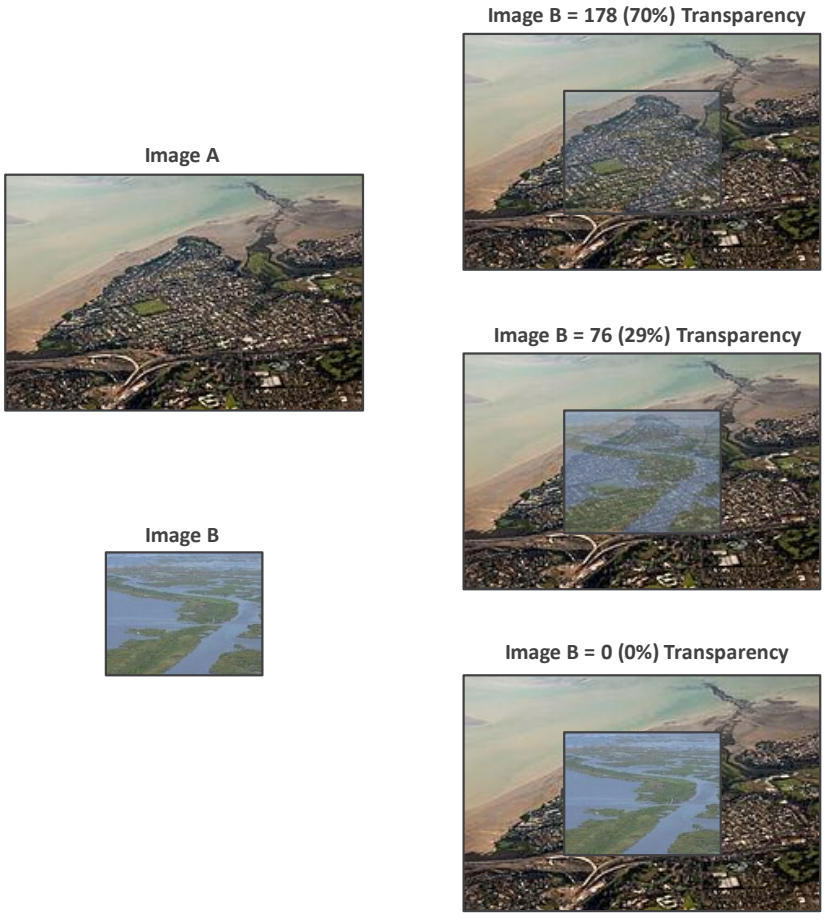


Figure 8: Images A and B superimposed using Transparency parameter

A transparency value of zero (0) for Image B represents fully opaqueness, where image A is not visible and image B is fully visible. As the transparency value increases to 76 (29%) to 178 (70%), Image A becomes more visible. A transparency value of 255 represents full transparency, where image B is not visible and image A is fully visible. Values between zero and 255 represent

levels of transparency for blending two images together. The default value for transparency is zero (0) when not included in the Local Set.

7.3.7 Z-Order

Z-Order (Item 18) defines the order of images (or stack) along the Z-axis (the axis perpendicular to the horizontal (left/right) axis, and the vertical (up/down) axis. A reserved value of zero (0) defines the bottom-most image in the stack. Different images cannot have the same z-order.

Requirement(s)	
ST 1602-04	Each Sub-Image within a Composite Image shall have a unique non-zero Z-Order value.
ST 1602.1-09	A Composite Imaging Local Set shall contain the Z-Order (Item 18) metadata item.

8 Usage of ST 1607

8.1 MIIS Core Identifier

Where there are multiple sensors within a Motion Imagery stream, the metadata within a segment is related to its parent through the MISB ST 1204 [7] MIIS Core Identifier. Thus, adding a Core Identifier is mandatory for use of more than one sensor per stream.

Requirement	
ST 1602.1-10	Where more than one sensor's metadata is included within a Motion Imagery stream, its MIIS Core Identifier in accordance with MISB ST 1204 shall be present within every child segment.

8.2 Metadata Substream Identifier (MSID)

A Metadata Substream Identifier (MSID) is a unique identifier within a Segment LS. This allows for multiple instances of Segment LSs within a metadata stream in support of sub-streams of sub-streams and Report-on-change. See the Motion Imagery Handbook for an in-depth discussion of sub-stream construction and the utility of the MSID. MISB ST 0601 defines requirements for the use of the MSID.

An instantiating metadata set defines a MSID metadata item and its requirements for its use. Every Segment LS implemented within that instantiating metadata set then includes that MSID as a metadata item.

9 Appendix A – Example: Four source images composited into one image - Informative

In this example, four independent sensors forming four Source Images are composited into one Composite Image as shown in Figure 9.

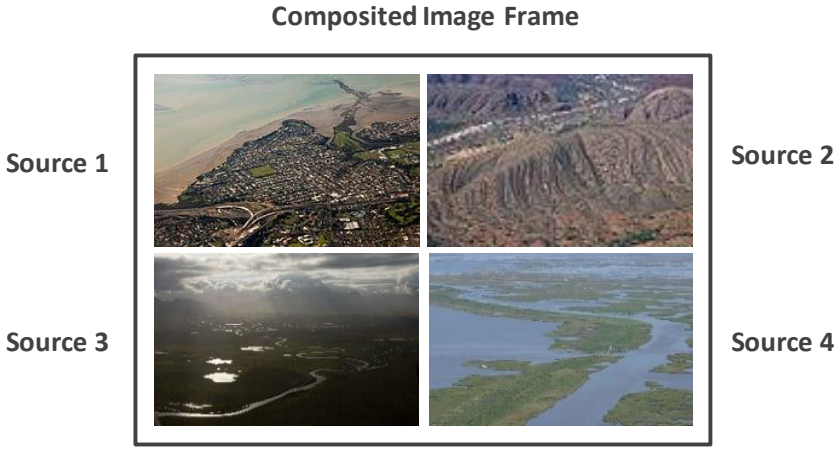


Figure 9: Four Source Images composited into one Composite Image

The MISB ST 0902 [8] metadata items are mandatory within Motion Imagery. In an image composed of four images, some items are common across the four images, whereas some items are specific to each image themselves. For example, the need for metadata to describe the geometry of each image, such as location of each image within the composite, the pixel density of each image, etc. The Composite Imaging LS provides this information.

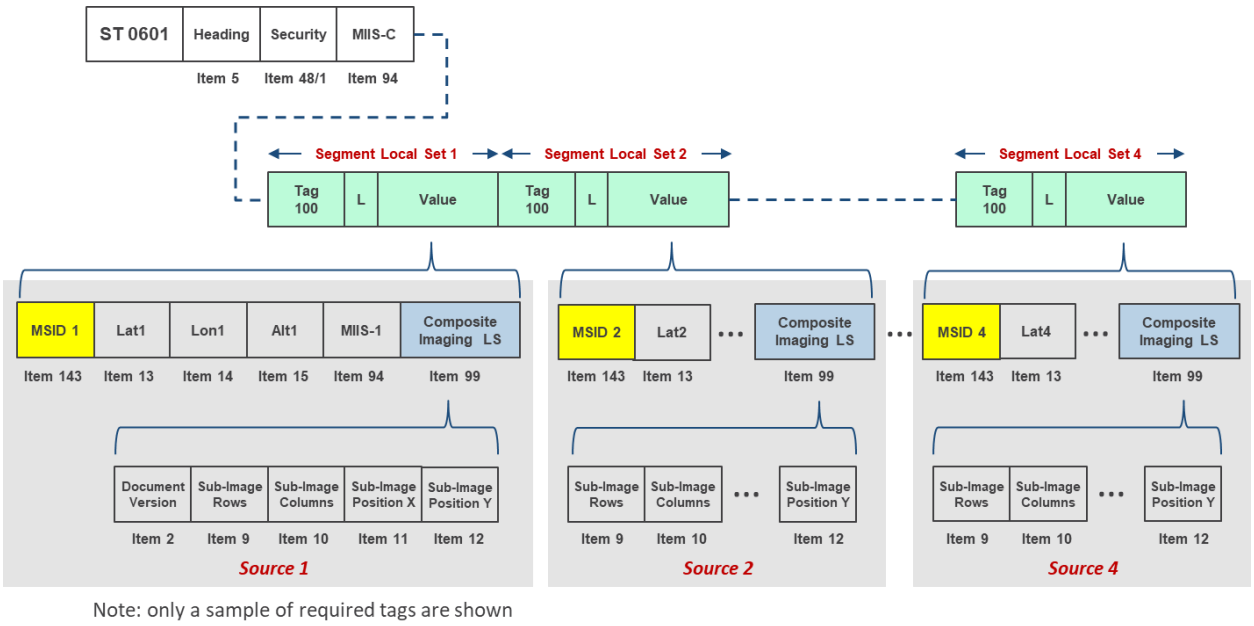
The Segment LS (Item 100) in MISB ST 0601 enables reuse of those metadata items from the ST 0601 LS unique to a specific image. Because there are four source images, there are four instances of Item 100. Each Segment LS must contain a MSID (see Req -11). ST 0601 defines a MSID (Item 143), which becomes one item in the Value of each Segment LS. Other items from ST 0601 likewise become items in the Segment LS Value portion as shown in Figure 10.

Note the reuse of ST 0601 Item 13/Lat, Item 14/Lon, and Item 15/Alt within Item 100 as they describe the specific attributes of each sensor. Each sensor has its own Motion Imagery Core Identifier Item 94/MIIS also inserted within the value of Item 100 as well. Finally, the Composite Imaging LS (Item 99) contains specific attributes for each source image.

This standard allows reuse of many ST 0601 items within a Segment LS. Exceptions include some items of the Security Local Set Tag 48 (see MISB ST 1607 for these exceptions). While there is no requirement that any ST 0601 items be present; the MISP [9] mandates a Motion Imagery stream contain all ST 0902 items. It is likely that some common ST 0902 items (i.e., mission, security, etc.) will be present at the ST 0601 parent level, and others, which vary per sensor source, will be present at the Segment LS Item 100 child level.

Note in this example the need for five Motion Imagery Core Identifiers (MIIS): one defining the composite image (i.e., MIIS-C) and one for each of the four sub-images (i.e., MIIS 1, MIIS 2, etc.).

See MISB ST 1607 for additional details on the use of the Segment LS construct.



Note: only a sample of required tags are shown

Figure 10: Use of Tag 100 – Embedding a Composite Imaging Local Set

10 Appendix B – Example: Two source images composited into one blended image - Informative

In this example, two independent sensors forming two Source Images blend into one Composite Image as shown in Figure 11.



Figure 11: Two Source Images blended

The MISB ST 0902 metadata items are mandatory within Motion Imagery. In an image composed of two images, some items are common across the two images, whereas some items are specific to each image themselves. For instance, assuming both images have similar dimensions and location in the composite, specific metadata describes the transparency of each image. Transparency (Tag 17) in the Composite Imaging LS provides this information, which is present for each image.

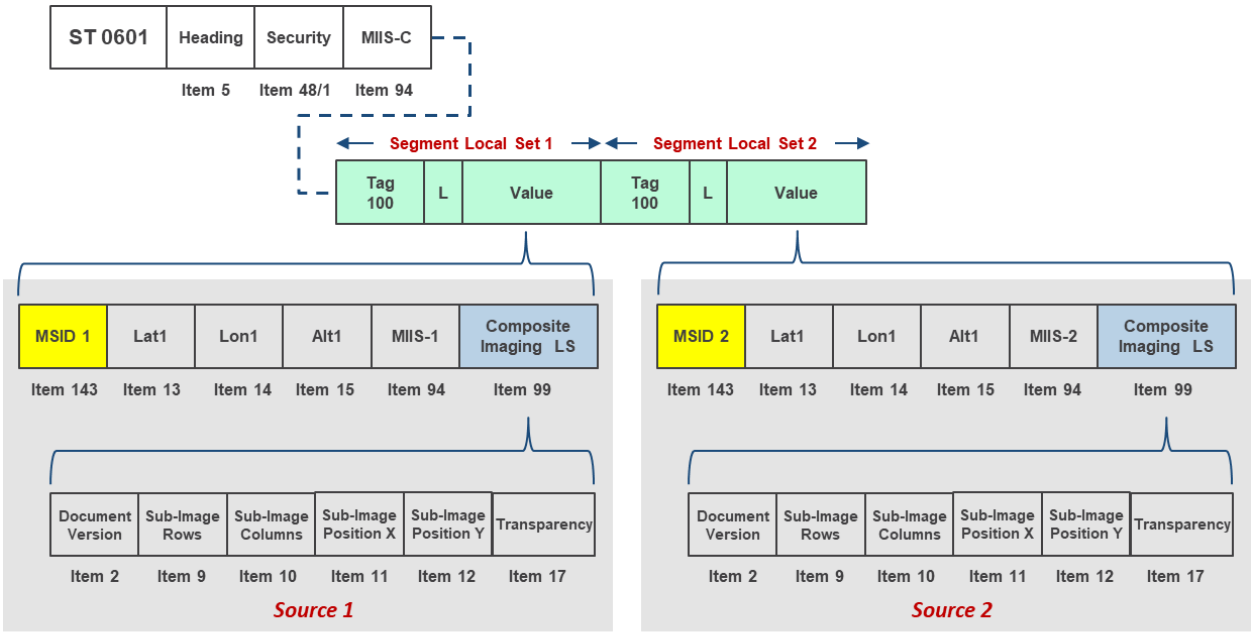
Each image in this example reuses four MISB ST 0601 tags. The Segment LS (Tag 100) in MISB ST 0601 enables the reuse of tags defined in a Local Set. Because there are two source images, each have a Tag 100. Within the Value of Tag 100, a Composite Imaging LS Tag 99 follows the four ST 0601 items specific to each source image as shown in Figure 12.

Note the reuse of ST 0601 metadata Tag13/Lat, Tag 14/Lon, and Tag 15/Alt within Tag 100 as these items describe the specific attributes of each sensor. Each sensor will have its own Motion Imagery Core Identifier (MIIS/Tag 94) carried within Tag 100 as well. Finally, the Tag 99 Composite Imaging LS includes the specific attributes of each source image.

Some “common” MISB ST 0601 tags (i.e., mission, security, etc.) will be present at the parent level, and others, which vary per sensor source, will be present at the Tag 100 child level.

Note in this example the need for three Motion Imagery Core Identifiers (MIIS): one defining the composite image (i.e., MIIS-C) and one for each of the four sub-images (i.e., MIIS 1, MIIS 2).

See MISB ST 1607 for additional details on the use of the Segment LS construct.



Note: only a sample of required tags are shown

Figure 12: Two Source Images blended

11 Appendix C – Example: One source image with picture-in-picture composited into one image - Informative

In this example, one sensor provides the information for two image views: the main sensor Source Image and a picture-in-picture (PIP) zoomed view of an area-of-interest of the Source Image as shown in Figure 13.



Figure 13: One Source Image with PIP composited into one Composite Image

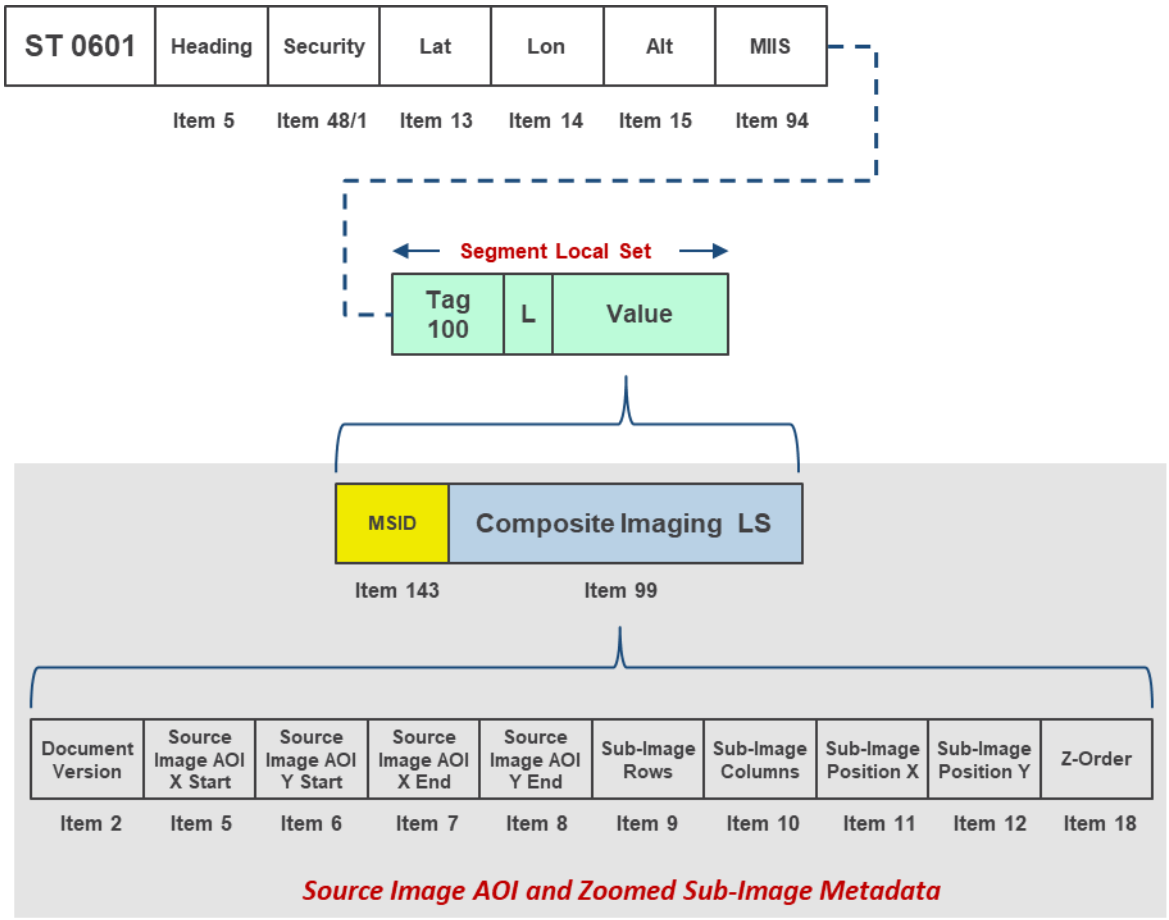
In this case, since both images emanate from the same source, all MISB ST 0902 metadata items are common to both. A Composite Imaging LS describes the position/dimensions of the AOI extracted, and the position/dimensions of the zoomed Sub-Image.

It is likely that the Source Image AOI may change during the Motion Imagery sequence, the zoom factor varied, or the zoom turned off. The temporal capability of this feature requires using the Segment LS (Tag 100) from MISB ST 0601. As changes as those mentioned occur, the information described by Tag 100 will change as well; that is, the Composite Imaging LS contained within the value of Tag 100 will have differing metadata values.

MISB ST 1202 [10] describes image transformations such as zoom. Once the zoomed image is available and ready for compositing with the source image, the Composite Imaging LS provides the information for position/dimensions of both the Source Image AOI and the zoomed Sub-Image result as shown in Figure 14.

In this example because the Source Image AOI and its derivative zoomed Sub-Image both emanate from the same source image, there is need for only one Motion Imagery Core Identifier (MIIS) in the metadata stream.

MISB ST 1607 and the Motion Imagery Handbook provide additional details on the use of the Segment LS construct.



Note: only a sample of required tags are shown

Figure 14: One Source Image with PIP zoomed Sub-Image overlay